

A Details of Dataset

A.1 Dataset Construction Prompts

Below is our prompt for querying GPT-4o to generate the options for Multi-Choice questions of each concept:

```
Please provide four answer options (A, B, C, D) for the following
question, and indicate the correct answer. Example: Question: 'When
was Costa Coffee founded?' Options: A) 1971 B) 1985 C) 1992 D) 2000
Correct Answer: A) 1971
Now, please answer the following question: Question: Question
Options:
```

Below is our prompt for querying the model to generate the answers for Multi-choice questions in three-shot setting:

```
**Question:** What is the capital city of France? **Options:** A.
Berlin B. Madrid C. Paris D. Rome **Answer:** C
**Question:** What is the largest planet in our solar system?
**Options:** A. Earth B. Jupiter C. Mars D. Venus **Answer:** B
**Question:** Which element has the chemical symbol "O"? **Options:**
A. Oxygen B. Gold C. Silver D. Iron **Answer:** A
**Question:** question **Options:** A. option a B. option b C. option
c D. option d
```

Below is our prompt for collecting the coefficients in model when querying about concept-related knowledge:

```
Question: question Answer: answer:
```

A.2 Manual Verification

Here, we describe the manual verification process used in constructing SpecWiki, including the validation of model-generated data:

Specifically, we analyze a subset of 524 (10%) questions from SpecWiki, by sampling 50% of the concepts and randomly selecting 2 questions per concept. Then, we manually verify that the questions are about the given concept and that they are simple and reasonable. In addition, we review all the generated questions for 200 sampled concepts and verify they are not repetitive. We find that all analyzed questions were about the given concept and that 522 (99%) of them are reasonable simple questions. Moreover, we observe that questions are generally diverse, with only 1 out of 20 concepts having 2 (out of 10) similar questions. This shows that our data generation process produces valid and diverse instances for evaluation.

A.3 Dataset Categories and Examples

Here, we provided a more detailed distribution of the categories and the corresponding example data of SpecWiki dataset in Table 4 and Table 5, respectively.

A.4 Validation of Popularity-Frequency Correlation

We included a simple experiment to validate the strong correlation between the popularity of concepts and their frequency in the pretraining data. To be specific, we used The Pile (Gao et al., 2020), which currently serves as a significant portion of the pretraining dataset for most large language models (Touvron et al., 2023a; Groeneveld et al., 2024; Team et al., 2024a), as an example of a pretraining corpus. We then counted the frequency with which each of the 525 concepts from SpecWiki appeared in all text segments of The Pile dataset via the Elasticsearch API (Elazar et al., 2024). Subsequently, we compared these frequencies with the popularity metrics for each concept and

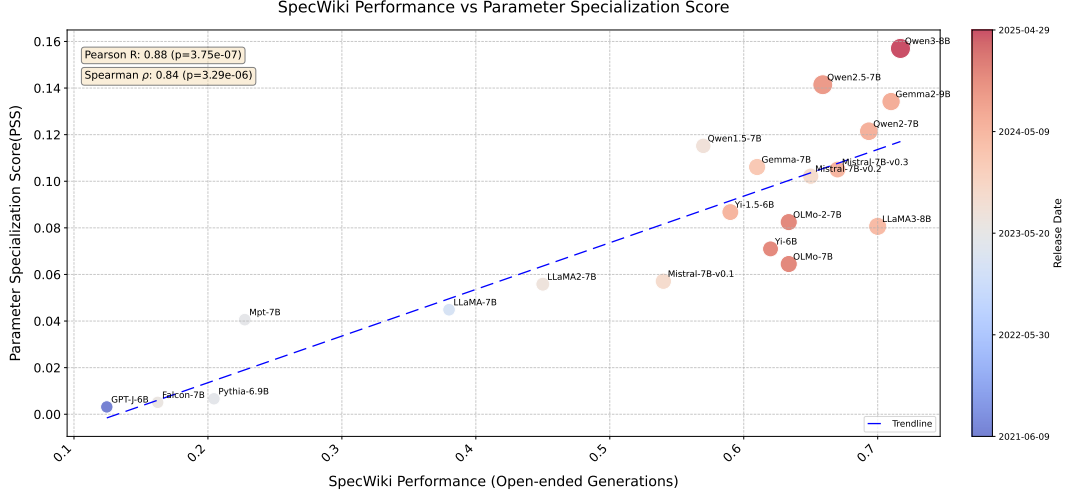


Figure 6: Performance across 20 models on Parameter Specialization Scores on Open-ended Generation Setting.

computed the corresponding Spearman’s rank correlation coefficient. The result is 0.814, indicating a strong correlation.

Additionally, Table 3 presents the top 3 most popular and the bottom 3 least popular concept examples in SpecWiki, along with their occurrence counts in The Pile dataset and their corresponding popularity scores.

Top 3 High Popularity Example Concept	Frequency	Popularity
Wikipedia	911708	1414686
Barack Obama	984586	1128538
India	3839241	1024513
Bottom 3 Low Popularity Example Concept	Frequency	Popularity
Dark Souls (video game)	49	27
Culture of Latin America	251	90
Array (data structure)	1164	94

Table 3: Top and bottom 3 concepts ranked by popularity and their corresponding frequencies.

B Details of Experiments

B.1 The implementation of the models

For all models, the inference is performed in a text completion/generation mode, without the addition of any instruction tokens, to better assess the knowledge present in the model. For the Multi-Choice Questions task, we use a three-shot setup for each model and search for the answer within the next 30 tokens generated by the model. For the open-ended generation task, we prompt the model in a zero-shot setting to produce an answer no longer than 150 tokens.

All the experiments in this work were conducted on four 80GB NVIDIA A800 GPUs.

B.2 Parameter Specialization Scores on OEG Setting

In Figure 6 we provided the performance across 20 models on Parameter Specialization Scores on Open-ended Generation Setting.

High Frequency (Number of Concepts: 181)				Medium Frequency (Number of Concepts: 191)				Low Frequency (Number of Concepts: 153)			
Country	13.3%	Technology	7.6%	Technology	19.9%	Mathematics	4.4%	Person	21.9%	Brand/Product	6.3%
Culture	9.5%	Brand/Product	7.6%	Art and Entertainment	11.1%	Politics	4.4%	History	10.6%	Medical	5.5%
Location	8.6%	Person	6.7%	Natural Sciences	10.5%	Location	4.4%	Entertainment	8.6%	Culture	2.9%
History	8.6%	Medical	6.7%	Medical/Biology	7.7%	Country	3.9%	Company/Organization	7.3%	Others	2.3%
Sports	7.6%	Entertainment	6.7%	Culture	7.2%	Company/Organization	3.3%	Others	6.3%	Natural Sciences	2.1%

Table 4: Ten most frequent concept categories of SpecWiki in high frequency, medium frequency, and low frequency levels.

Example Concept	Frequency Level	Category	Example Multi-Choice QA	Example Open-ended Generation
The Lord of the Rings	High Monthly Views: 177540	Art and Entertainment	Question: "Who is the main protagonist of 'The Lord of the Rings'?" Options: A: "Frodo Baggins", B: "Gandalf the Grey", C: "Aragorn", D: "Legolas" Answer: A	Question: "Who is the author of 'The Lord of the Rings' trilogy?" Answer: "J.R.R. Tolkien."
Detritivore	Medium Monthly Views: 11810	Biology	Question: "What do detritivores consume to obtain nutrients?" Options: A: "Fresh, living plants and animals", B: "Detritus, including decomposing plant and animal parts and feces", C: "Sunlight and water", D: "Inorganic minerals and metals" Answer: B	Question: "What term is used for the consumption of dead wood by detritivores?" Answer: "Sapro-xylophagy."
Maluma	Low Monthly Views: 2252	Person	Question: "In which city was Maluma born and raised?" Options: A: "Bogotá", B: "Cali", C: "Medellín", D: "Cartagena" Answer: C	Question: "What is the name of Maluma's 2023 album?" Answer: "Don Juan"

Table 5: Example data from the SpecWiki dataset.

B.3 Hallucination Metric Descriptions

In this experiment, we additionally incorporate two metrics, Semantic Entropy (Kuhn et al., 2022) and Local Intrinsic Dimension (LID) (Yin et al., 2024), to assess hallucination. This helps evaluate whether the finetuning methods that enhance Parameter Specialization also effectively mitigate the unintended side effect of hallucination in the model’s output.

Semantic Entropy Semantic entropy is defined as a measure of uncertainty based on the distribution of semantically equivalent outputs. In this method, the outputs are grouped into clusters of semantically similar responses, and the entropy is calculated among these groups. Formally, it is expressed as:

$$\text{Semantic Entropy} = \frac{1}{|C|} \sum_{i=1}^{|C|} \log p(C_i|x)$$

where C_i represents the summed likelihood of outputs in the i -th group, and $|C|$ is the total number of such groups. The measure captures the uncertainty not in individual responses but within clusters of semantically similar outputs. This approach accounts for semantic equivalence among different responses, providing a more robust evaluation of entropy in generative tasks.

Local Intrinsic Dimension The Local Intrinsic Dimension (LID) method detects hallucinations in Large Language Models by measuring the discrepancy in the local intrinsic dimension of model activations. This approach is grounded in the principle that LID represents the minimal number of activations required to characterize a data point, with truthful outputs exhibiting lower LID values due to their closer alignment with natural language structure, while hallucinated outputs tend to show higher LID values due to mixing human prompt and model distributions. Technically, the method employs Maximum Likelihood Estimation (MLE) using a Poisson process to approximate the count of neighbors surrounding sample points, computed through the formula $m(X_i) = (1/(T - 1) * \sum(\log(Q_T/Q_j)))^{-1}$, where T represents the number of nearest neighbors and Q_j denotes the Euclidean distance to the j -th nearest neighbor. For more details about the Local Intrinsic Dimension metric, please refer to the work (Yin et al., 2024).